Tests & Diagnostics — AGEL0206 σ_e ApJL paper

Last updated: 2026-05-04 Headline: σ_e(<R_e) = 268 ± 32 km/s (stat ±24 ⊕ I-shape ±13 ⊕ mask ±16 ⊕ frame ±5 ⊕ centering ±4) Method: §6cum cumulative I-weighted ppxf at R<R_e=2.305" inside the F200LP-masked, frame-aware, SPS-pooled (FSPS+EMILES+XSL) bootstrap × polynomial-degree posterior

This document is the canonical index of every test, sweep, audit, and sensitivity check run for the σ_e measurement. Each row points to the script/notebook that runs it and the result file or note it produced.

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0. Quick-reference test matrix

#	Test	Where	Result	Status
A. Foundational				
A1	KCWI cube provenance	`NOTES_methodology_2026-04-27.md`	Multi-night Aug/Sept/Dec 2025; header mislabel only	✓
A2	Systemic redshift via line fitting	`notebooks/04_redshift_verification.ipynb`, `scripts/redshift_verify.py`	z = 0.67564 (vs DR2 0.67511)	✓
A3	R_e from masked curve-of-growth (F140W+F200LP)	`scripts/measure_Re.py`, `scripts/final_sigma_e.py:curve_of_growth`	R_e = 2.305" = 16.23 kpc	✓
A4	HST-mean center via centroid_2dg	`scripts/final_sigma_e.py:find_center`	F140W & F200LP agree to 0.36"	✓
A5	Bagpipes SED fitting (M⋆) — aperture vs Sersic-total	`notebooks/02_streamlined_Bagpipes_SED.ipynb`, `08_sersic_total_photometry.ipynb`	log M⋆ = 11.33 +0.07/—0.09 (aperture, headline); 11.40 +0.11/—0.15 (Sersic-total, +0.065 dex / +16%)	✓
B. Pipeline correctness audits				
B1	Instrumental LSF audit (DISPSCAL=0.294)	`scripts/ifu_spectral_resolution.py`	FWHM=0.692 Å; σ_v_inst≈12-14 km/s	✓
B2	σ_inst sensitivity (×0.5 to ×2.0 LSF)	`scripts/sigma_inst_sensitivity.py`, audit 3	max |Δσ| = 0.83 km/s	✓
B3	SPS template wavelength frames	`scripts/audit_ppxf_methodology.py`, NOTES Test 2	FSPS=vacuum, EMILES=air, XSL=air (Ca K minimum + V_sys)	✓
B4	V_sys air vs vacuum × 5 polynomial degrees	`scripts/audit_ppxf_methodology.py` audit 1	ΔV consistent within ±2 km/s across degrees	✓
B5	z × air-vac differential (obs vs rest application)	`scripts/audit_ppxf_methodology.py` audit 2	1.83 km/s differential — sub-budget	✓
B6	fwhm_gal_dict frame consistency check	`scripts/audit_ppxf_methodology.py` audit 4	dict in REST frame, matches Cappellari pattern	✓
C. SPS templates & frame-fix				
C1	Frame-aware ppxf: per-SPS native frame	`scripts/bootstrap_ppxf.py:SPS_NATIVE_FRAME` + `frame_galaxy='auto'`	V_sys split collapsed from ~110 to ~15 km/s	✓
C2	End-to-end V_sys closure (frame swap)	NOTES Test 3, addendum 2026-04-29	Frame fix matches diagnosis	✓
C3	σ shift from frame fix	NOTES addendum, `error_budget()`	≤5 km/s across SPS → carried as ±5 km/s budget	✓
C4	3-SPS pooled posterior	`scripts/final_sigma_e.py:pool_sps`	Per-SPS V_sys subtracted before pooling	✓
D. Aperture & I(r)				
D1	§6cum cumulative I-weighted ppxf	`scripts/final_sigma_e.py:run_aperture_sps`	σ_e(<R_e) = 267.82 km/s headline	✓

#	Test	Where	Result	Status
D2	§6cum vs §7 (annular) cross-check	`notebooks/07c`, `~/.claude/.../reference_cumulative_vs_annular_sigma_e.md`	§7=255, §7b=271 — all <1σ from §6cum	✓
D3	I(r) shape sweep (10 shapes, fixed mask)	`scripts/run_isource_shape_sweep.py`, `scripts/analyze_isource_shape_sweep.py`	Range = 12.9 km/s (excluding F200LP_Sersic2D outlier) → ±13 budget	✓
D4	I(r) shape sweep refresh post-frame-fix at N=500	NOTES addendum (d), 2026-04-29	Frame fix has <0.5 km/s impact; ±13 budget validated	✓
D5	Aperture choice: R<R_e/8, R<R_e/2, R<R_e	`scripts/final_sigma_e.py:APERTURE_FRACS`	R<R_e/8 dropped (inside seeing FWHM/2 = 0.64")	✓
D6	Equal-N vs equal-width annular binning	`notebooks/07b` (5-bin), `notebooks/07c` (equal-N)	Equal-N inside R_safe=3R_e/4 chosen	✓
D7	R_e source systematic (mean vs F140W vs F200LP vs CaHK+G)	`scripts/final_sigma_e.py` §7, `notebooks/09 §7b`	Spread = 16.9 km/s (mean=268, F140W=265, F200LP=276, CaHK=282); sub-budget	✓
E. Mask treatment				
E1	F200LP arc mask reprojected to IFU grid	`scripts/final_sigma_e.py:load_setup`	0.7% of all spaxels flagged (~38 inside R<R_e)	✓
E2	Hard mask (w=0.0) headline	`scripts/final_sigma_e.py`, track A	σ_e = 267.82 ± 24 km/s	✓
E3	No-mask sensitivity (w=1.0)	track B	σ_e = 252.82, Δ = −15.0 km/s	✓
E4	Soft-mask single-point (w=0.5)	`scripts/soft_mask_track.py`, track C, NOTES addendum (b)	σ_e = 258.54, Δ = −9.29; super-linear	✓
E5	5-point mask-weight sweep w∈{0,0.25,0.5,0.75,1}	`scripts/soft_mask_track.py --weight ...`, `scripts/analyze_mask_weight_sweep.py`, NOTES addendum (c)	Per-step drops 5.0→4.3→3.5→2.2; quadratic c=+7.3 (concave-up); threshold-dominated	✓
E6	Mask dilation (over-masking diagnostic)	`notebooks/07d_sigma_e_forceful_mask.ipynb`, `~/.claude/.../project_nb07d_overmasking_finding.md`	Dilation inflates σ via noise; do NOT dilate (negative finding)	✓
E7	Arc-spectrum subtraction sibling	`notebooks/07e_sigma_e_arc_subtract.ipynb`	Matches §6cum within 0.1 km/s — residual arc dilution sub-dominant	✓
E8	Arc-as-sky mechanism test (no-mask + arc-sky template)	`scripts/run_07f_arc_sky.py`, `notebooks/07f_arc_sky_subtract.ipynb`	σ_D=274.6 vs σ_A=267.8 / σ_B=252.8; recovery=145% — dilution explains the full no-mask Δ	✓
F. Centering				
F1	5-center sweep (±0.4" perturbations)	`NOTES_centering_investigation_2026-04-27.md`, `scripts/regen_s6cum_nomask_diagnostic.py`	σ_e spread = 3.7 km/s → ±4 km/s budget	✓
F2	HST F140W vs F200LP centroid offset	`scripts/final_sigma_e.py:find_center`	0.36" — driven by F200LP arc; HST-mean robust	✓
G. Bootstrap & polynomial degree				
G1	Wild bootstrap (Rademacher × local residual)	`scripts/bootstrap_ppxf.py:run_bootstrap_single_degree`	75-pix rolling window for σ_loc	✓

#	Test	Where	Result	Status
G2	Parallel bootstrap (joblib, BLAS=1 per worker)	`scripts/bootstrap_ppxf_parallel.py`	~6 min for 6 fits at N=500	✓
G3	Polynomial degree sweep (deg 15-29)	`notebooks/03c`, `scripts/build_nb09.py:§6.5`	σ flat within bootstrap envelope; polynomial saturated	✓
G4	N=50 smoke vs N=500 production agreement	`scripts/final_sigma_e.py --n_bootstrap`	Within 1 km/s	✓
H	Cross-checks (independent estimators)			
H1	§7 discrete Gültekin annular sum (frame-aware, refresh 2026-05-01)	`scripts/refresh_07c_gultekin.py`, `notebooks/07c §7`	256.17 −13.0/+12.7 km/s (post-fix; was 254.99 −24/+28 pre-fix)	✓
H2	§7b flat-σ extrapolation (frame-aware, refresh 2026-05-01)	`scripts/refresh_07c_gultekin.py`, `notebooks/07c §7b`	274.37 −16.2/+17.4 km/s (post-fix; was 271 −33/+35 pre-fix)	✓
H3	F200 mask sensitivity at N=500	`results/annular_bootstrap_07c_nomask/`	Δ_mask = −16.4 km/s → $\pm$16 budget	✓
I. Earlier sigma-discrepancy work (Tests 1-21)				
I1	NB01 vs NB05 σ reproduction	`notebooks/05x` Tests 1-4	Confirmed σ depends on aperture	✓
I2	Source mask + PSF-aware masking	`notebooks/05x` Tests 7-10	F140W mask flags deflector core; F200LP mask is the right arc mask	✓
I3	Threshold sweep + contamination weighting	`notebooks/05x` Tests 11-13	Pre-Gültekin masking strategy work	✓
I4	V_rms profile + bootstrap	`notebooks/05x` Tests 14-16	Pre-Gültekin radial work	✓
I5	Redshift sensitivity (z=0.67511 vs 0.67564)	`notebooks/05x` Test 17	<1 km/s effect on σ	✓
I6	Voronoi binning attempt	`notebooks/05x` Test 18	Failed — abandoned	✓
I7	PowerBin spatial binning	`notebooks/05x` Test 19, `scripts/run_powerbin_test19.py`	Rejected as binning scheme — outer-bin σ>800 km/s	✓
I8	R_e four-way comparison	`notebooks/05x` Test 20, `scripts/measure_Re.py`	F140W+F200LP masked CoG mean = 2.305” headline	✓
I9	Per-spaxel vs PowerBin rotation map	`notebooks/05x` Test 21	No rotation on Re-scale; σ-dominated	✓

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1. Headline numbers

σ_e at three apertures (cumulative I-weighted ppxf, frame-aware, masked)

Aperture	σ_e [km/s]	Notes
R<R_e (=2.305”)	268 $\pm$ 32 (stat $\pm$24)	paper headline — KH13/Greene+20 σ_e
R<R_e/2 (=1.152”)	224 $\pm$ 18	gradient diagnostic
R<R_e/8 (=0.288”)	dropped	inside seeing FWHM/2 = 0.64”

Error-budget composition (R<R_e)

Component	Value	Source
Statistical (bootstrap pooled 1σ)	$\pm$24	nb09, this paper

Component	Value	Source
I-shape spread (10 shapes excluding outlier)	±13	nb07c sweep, refreshed N=500 post-frame-fix
Mask on/off Δ	±16	nb07c, nb09 §5
Frame fix (max σ shift across SPS)	±5	NOTES addendum + audit 1
Centering (5-center sweep, ±0.4")	±4	`NOTES_centering_investigation_2026-04-27.md`
Quadrature total	±32	`scripts/final_sigma_e.py:error_budget`

Sensitivity values (for paper text)

- Soft-mask (w=0.5) at R<R_e: 258.5 km/s — Δ from headline = −9.3 km/s
- No-mask (w=1.0) at R<R_e: 252.8 km/s — Δ from headline = −15.0 km/s
- σ_e(w) is essentially linear with weak super-linearity (quadratic c=+7.3)

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2. Detailed test catalog

This expands each row of the matrix above. Subsections reference scripts, notebooks, and result files — open the linked file for the detailed code and the inline plots/output.

A. Foundational data and geometry

A1 KCWI cube provenance — The cube `Nov17_2025_DESJ0206_RL_combined_icubes_wcs.fits` is the final multi-night reduction (Aug 29 K409 + Sept 29 K409 + Dec 29 + Nov 17 2025). The header is mislabeled (DATE-OBS/PROGNAME/PROGPI describe only the last stacking pass). Settled in `NOTES_methodology_2026-04-27.md`. Cite the multi-night provenance, not the header.

A2 Systemic redshift — Independent line-by-line Gaussian fitting of Ca H+K, [OII], G-band, etc. gives z_systemic = 0.67564 (NIST air wavelengths). DR2 catalog z=0.67511 (used by older notebooks 03/03b) is offset by ~95 km/s from the line-fit value; nb09 uses 0.67564 throughout. Notebook: `notebooks/04_redshift_verification.ipynb` Module: `scripts/redshift_verify.py` (provides ABSORPTION_LINES / EMISSION_LINES dicts, `gaussian_fit`, `multi_line_fit`, `oii_doublet_fit`)

A3 Effective radius R_e = 2.305“ — Mean of F140W and F200LP masked curve-of-growth values (2.168 and 2.441”). The masked CoG zeros mask-flagged HST pixels before the radial integral so the arc/companion contributions don’t bias the bright-end half-light point. Computed in `scripts/measure_Re.py` and re-derived in `scripts/final_sigma_e.py:curve_of_growth` for the nb09 pipeline. Cross-check: IFU white-light proper-masked CoG gave 2.61” earlier — the HST-based 2.305” is now the headline.

A4 HST-mean center — `photutils.centroid_2dg` on F140W and F200LP cutouts (with their own `_mask.fits` as the mask) gives sub-pixel centroids in each HST WCS. We average in world coordinates, then propagate to IFU sub-pixel: `scripts/final_sigma_e.py:find_center`. F140W vs F200LP centroid offset = 0.36” — flags the F200LP arc-affected detection but the HST-mean is robust (verified by the 5-center sweep, F1).

A5 Stellar mass — Bagpipes SED fitting on HST + JWST aperture photometry (notebook 02, N_posterior = 500). Bagpipes config: exponential-τ SFH, BPASS+CLOUDY templates, Calzetti dust, free metallicity, redshift fixed at 0.67564.

Per-filter observed flux + best-fit modelled flux:

Filter	λ_pivot [Å]	AB obs	AB model (p50)	F_obs [10^{-18} cgs]	F_model (p50)	residual
F200LP	4 972	22.613	22.563	3.97 ± 0.06	4.16	+4.7%
F140W	13 923	19.134	19.264	12.47 ± 0.05	11.06	−11.3%
F150W2	16 720	18.942	19.033	10.31 ± 0.005	9.49	−8.0%
F322W2	32 470	18.604	18.505	3.73 ± 0.001	4.09	+9.6%

Posteriors:

Parameter	p16	p50	p84
log $M_\star/M_\odot$	11.24	11.33	11.41
mass-weighted age [Gyr]	3.69	5.10	6.11
exponential SFH age [Gyr]	4.46	5.98	6.97
exponential τ [Gyr]	0.43	0.68	1.21
metallicity $Z/Z_\odot$	0.17	0.89	1.87

Parameter	p16	p50	p84
A_V [mag]	0.34	0.82	1.55
z_phot	0.674	0.675	0.676

$\rightarrow$ M⋆ = 2.15 × 10^{11} M⊙; z_phot agrees with the line-fit z = 0.67564 within ±0.001.

The model under-predicts the two NIR bands (F140W, F150W2) by ~8-11% and over-predicts the bluest+reddest by ~5-10%. Classic "redder NIR slope than a single-τ SFH prefers" tension; not catastrophic, but flags that a more flexible SFH (delayed-τ or non-parametric) might tighten the fit if revisited.

Sersic-total cross-check (`notebooks/08_sersic_total_photometry.ipynb`, `scripts/measure_Re.py`-style 2D Sersic fits per filter, extrapolated to total flux):

Filter	AB aperture	AB Sersic-total	Δmag	Sersic n	r_eff [″]
F200LP	22.613	20.672	−1.94	1.42	2.49
F140W	19.134	18.282	−0.85	1.54	1.88
F150W2	18.942	18.154	−0.79	1.40	1.72
F322W2	18.604	17.633	−0.97	1.97	2.03

Re-fed through Bagpipes with the same priors:

Quantity	aperture (paper)	Sersic-total	Δ
log M⋆/M⊙ p50	11.33 +0.07/−0.09	11.40 +0.11/−0.15	+0.065 dex
M⋆ [M⊙]	2.15 × 10^{11}	2.49 × 10^{11}	+16%

The aperture under-counts outer-profile flux by ~0.8-1.9 mag depending on filter, but at the integrated-mass level the Sersic correction is +0.07 dex — within the aperture-photometry quoted uncertainties. The paper cites the aperture value as the headline; Sersic-total is the sensitivity floor.

Cache: `results/bagpipes_sed_results.npz` (aperture, 500-sample posterior), `results/bagpipes_sersic_refit.npz` (Sersic-total), `results/sersic_total_photometry.npz` (Sersic fit parameters). Figure: `results/AGEL0206_spectra_SED_fit.pdf`.

B. Pipeline correctness audits (**`scripts/audit_ppxf_methodology.py`**)

Four orthogonal correctness audits run end-to-end on the headline aperture spectrum to verify ppxf usage matches Cappellari (2017, 2023). Saved in `results/ppxf_methodology_audit.npz`. Documented in `NOTES_nb09_frame_fix_and_final_sigma_e_2026-04-28.md` ADDENDUM 2026-04-29.

B1 Instrumental LSF — KCWI header DISPSCAL = 0.294 (instrumental σ in pixels) × CD3_3 = 1.000 Å/pix $\rightarrow$ FWHM_inst = 2.355 × 0.294 = 0.692 Å (constant in observed Å). At the rest-frame fit window (3879–4476 Å), σ_v_inst ≈ 12–14 km/s. Galaxy $\sigma \approx$ 270 km/s is resolved by ~20×. Diagnosed in `scripts/ifu_spectral_resolution.py`.

B2 σ_inst sensitivity — Sweep DISPSCAL × {0.5, 0.75, 1.0, 1.25, 1.5, 2.0}. Max $|\Delta\sigma|$ = 0.83 km/s across all SPS. Reason: ppxf's `sps_lib` clips fwhm_diff^2 = (FWHM_gal^2 − FWHM_tem^2).clip(0) (sps_util.py:169). When templates are intrinsically broader (FSPS, EMILES at our band), no convolution is applied — σ becomes insensitive to DISPSCAL. LSF is rock-solid. Script: `scripts/sigma_inst_sensitivity.py`

B3 SPS template frames — Located the Ca K minimum in each shipped template: - FSPS spectra_fsps_9.0.npz: Ca K at 3934.86 Å $\rightarrow$ +1.20 Å vs air rest = vacuum - EMILES spectra_emiles_9.0.npz: 3933.82 Å $\rightarrow$ +0.16 Å = air - XSL spectra_xsl_9.0.npz: V_sys closes at +9 km/s with air-conversion $\rightarrow$ air End-to-end V_sys closure (NOTES Test 3) confirms.

B4 V_sys air vs vacuum × 5 polynomial degrees — Run ppxf at degrees {15, 18, 21, 24, 27} for each SPS in both frames. ΔV(air − vac) ≈ +83 km/s for FSPS, ≈ −82 km/s for EMILES & XSL, consistent within ±2 km/s across degrees. Frame identification is robust.

B5 z × air-vac differential — Apply `vac_to_air` at observed wavelengths (KCWI 6500–7500 Å) $\rightarrow$ v_offset = −82.7 km/s. Apply at rest wavelengths (3879–4476 Å) $\rightarrow$ −84.5 km/s. Differential = 1.83 km/s — sub-budget. Either convention is defensible; we follow Cappellari's pattern (apply at obs).

B6 fwhm_gal_dict frame check — Confirmed `_prep_spectrum_for_ppxf` builds the dict as `{"lam": lam_gal_rest, "fwhm": fwhm_gal_rest}` in REST frame, matching `ppxf_example_high_redshift.py:99–101`. FWHM_obs = 0.692 Å → FWHM_rest = 0.413 Å at z=0.67564. `sps_util.py:167` interpolates this onto template `lam_temp` (also rest), then `varsmooth` pre-broadens.

C. SPS templates and frame-fix

C1 Frame-aware ppxf — `scripts/bootstrap_ppxf.py` exposes `frame_galaxy='auto'` which reads `SPS_NATIVE_FRAME = {fsps: vacuum, emiles: air, xsl: air}`. The galaxy spectrum is prepared in the matching frame (`util.vac_to_air()` applied as scalar median when needed, matching `ppxf_example_kinematics_sdss.py` :127-134). Diagnosed Apr 2026 — pre-fix the pipeline always converted to air, producing a −90 km/s FSPS V_sys offset.

C2 V_sys closure — Same galaxy spectrum, swap only the frame. Pre-fix: FSPS −90, EMILES +3, XSL +9. Post-fix (frame-aware): FSPS −7, EMILES +3, XSL +9. Split-track collapsed from ~110 → ~15 km/s. NOTES Test 3.

C3 σ shift from frame fix — Side-effect on σ: $\leq$5 km/s per SPS, $\leq$2 km/s on the 3-SPS pool. Carried as $\pm$5 km/s in the error budget (ironically dominated by the per-SPS shifts that pool out, but conservative).

C4 3-SPS pooled posterior — `scripts/final_sigma_e.py:pool_sps` concatenates per-SPS bootstrap σ samples after subtracting per-SPS V_sys medians (so V offsets don't inflate the V posterior; σ posteriors are concatenated raw — no V_sys correction needed for σ). The headline σ_e(<R_e) is the 16/50/84 percentile of this combined-SPS pool.

D. Aperture and I(r) framework

D1 §6cum cumulative I-weighted ppxf — For each spaxel inside R<R_max, build I-weight from the IFU 6500–7500 Å white-light band, drop arc-mask-flagged spaxels, normalize and sum to a single 1-D spectrum, fit ppxf. This is the headline path — matches KH13 / Greene+20 / SAURON / ATLAS3D / MaNGA σ_e definition (Cappellari+2006 eq. 1). Cross-references in `~/.claude/.../reference_cumulative_vs_annular_sigma_e.md`. Code: `scripts/final_sigma_e.py:run_aperture_sps`.

D2 §6cum vs §7 (annular) — §7 discrete Gültekin annular sum at R<R_safe gives 254.99 km/s; §7b flat-σ extrapolation gives 271 km/s; both within 1σ of §6cum's 267.82 km/s. §6cum is the headline because (a) it matches the literature definition, (b) it preserves LOSVD line-shape information that §7's moment-pooling discards, (c) bright-center I-weighting auto-suppresses arc contamination.

D3 I(r) shape sweep (10 shapes) — Hold the spaxel selection fixed (F200-raw mask, R<R_e), vary only the I-weight map. Shapes: 1. IFU_band — 6500-7500 Å (headline) 2. IFU_wl — full IFU white-light 3. F140W_raw — HST F140W reprojected 4. F200LP_raw — HST F200LP reprojected 5. F140W_arcmasked — F140W with arc zeroed 6. F200LP_arcmasked — F200LP with own arc-mask zeroed 7. F140W_1Dcog — annular-mean F140W → interp1d to spaxels 8. F200LP_1Dcog — annular-mean F200LP → interp1d 9. F140W_Sersic2D — full 2D Sersic fit 10. F200LP_Sersic2D — full 2D Sersic fit (poor fit, n=0.30 — outlier)

Range excluding F200LP_Sersic2D outlier = 12.9 km/s → $\pm$13 km/s budget. Script: `scripts/run_isource_shape_sweep.py` Analysis: `scripts/analyze_isource_shape_sweep.py` Caches: `results/annular_bootstrap_07c_ishape/{shape}_{sps}.npz` Figure: `results/figures/nb07c_ishape_sweep.png`

D4 I-shape sweep refresh post-frame-fix — Original sweep at N=250 predated the frame fix. Re-ran at N=500 with `frame_galaxy='auto'` on 2026-04-29. Per-SPS σ shifts <0.5 km/s; spread unchanged at 12.9 km/s; $\pm$13 budget validated. Pre-frame caches preserved at `results/annular_bootstrap_07c_ishape_oldframe/`. NOTES addendum (d).

D5 Aperture choice — Three apertures considered: R<R_e/8 (=0.288"), R<R_e/2 (=1.152"), R<R_e (=2.305"). R<R_e/8 dropped because it sits inside half the seeing FWHM (=0.64") — only 3 spaxels, and the inner σ would underestimate due to seeing-broadened sampling. R<R_e is the paper headline; R<R_e/2 is quoted as a gradient diagnostic.

D6 Equal-N vs equal-width annular binning — For the §7 cross-check only. Equal-N inside R_safe = 3R_e/4 + 1 outer flagged bin (`notebooks/07c`) gives balanced bootstrap variance per bin and isolates arc contamination in the outer bin. Equal-width annuli (`notebooks/07b`, 5 bins) shifted σ_e by ~10 km/s — at the SPS-systematic level ($\pm$24). Equal-N is the §7 paper choice.

D7 R_e source systematic (mean vs F140W vs F200LP vs Ca H+K + G) — The headline integrates the I-weighted aperture spectrum inside the mean F140W+F200LP masked CoG R_e (=2.305"). Three alternative R_e definitions test sensitivity to that choice (Track A masked, N=500):

R_e source	R_e ["]	σ_e [km/s]	$\Delta\sigma$_e
mean (paper)	2.305	267.82	—
F140W only	2.168	264.87	−2.96
F200LP only	2.441	275.86	+8.04
Ca H+K + G-band depth	2.866	281.74	+13.92

Spread = 16.9 km/s — at the mask-budget level (±16) but sub-budget vs the total ±32. σ_e rises monotonically with R_e (more outer-bulge spaxels in the I-weighted aperture). The Ca H+K + G-band absorption- depth I-map (rest 3925-3942, 3960-3976, 4297-4313 Å, summed `(continuum − flux)` per spaxel — see `cahk_g_line_depth_map` in `scripts/final_sigma_e.py`) is the most stellar-only of the four; its larger R_e reflects that the deflector light extends further than the F200LP UV-leaning band suggests. Code: `scripts/final_sigma_e.py` §7 + `cahk_g_line_depth_map`. Display: `notebooks/09 §7b` + `results/figures/nb09_re_source_systematic.png`. Caches: `results/final_sigma_e_paper/Re_{F140W,F200LP,CaHK}_{sps}_N500.npz`.

E. Mask treatment

Masking vs no-masking — pro/con summary

Track	Pro	Con
Hard mask (w=0.0, headline)	Removes arc contamination; matches literature convention; F200LP mask is purpose-built for this lens	Drops 38 of 184 spaxels inside R<R_e (~21% by count, ~27% by I-weight) $\rightarrow$ lower S/N; hard boundary may over-clip a few edge spaxels
No mask (w=1.0, sensitivity)	Maximum S/N; no boundary artifacts	Arc adds low-velocity continuum dilution $\rightarrow$ σ biased low by 15-17 km/s (Δ_mask = -16 km/s at R<R_e)
Soft mask (w=0.5)	Smooth middle ground	Super-linear in w (quadratic c=+7.3); threshold-dominated bias $\rightarrow$ first 25% of arc weight already captures 34% of total mask sensitivity. Arbitrary weight choice; not a literature convention
Mask dilation (E6)	—	Inflates σ via noise (dropped spaxels are S/N contributors, not bias contributors). Negative finding — do not dilate

The 16 km/s no-mask Δ is propagated into the error budget as `σ_mask = ±16`, so the choice is bounded, not assumed.

E1 F200LP arc mask reprojection — F200LP `_mask.fits` (HST 0.04“/pix, arc-tuned) is reprojected to the IFU 0.30”/spax grid via `scipy.ndimage.map_coordinates(order=0)` (nearest- neighbor). 69 of 10000 spaxels (0.69%) flagged. Inside R<R_e: 38 of 184 spaxels in the aperture, ~21% — but only ~27% of the I-weight (arc spaxels are mostly off-center).

E2 Hard-mask (w=0.0) headline — track A. E3 No-mask (w=1.0) sensitivity — track B. Δ_mask = -15.0 km/s. E4 Soft-mask single point (w=0.5) — track C. Δ_soft = -9.3 km/s. E5 5-point mask-weight sweep (w$\in${0,0.25,0.5,0.75,1}) — Per-step σ drops at R<R_e: 5.0 $\rightarrow$ 4.3 $\rightarrow$ 3.5 $\rightarrow$ 2.2 km/s. Quadratic c = +7.3 (concave-up, super-linear). Threshold-dominated bias: a few heavily-arc-contaminated spaxels carry most of the bias. First 25% of arc weight captures 34% of the total mask sensitivity. Scripts: `scripts/soft_mask_track.py`, `scripts/analyze_mask_weight_sweep.py` Figure: `results/figures/nb09_mask_weight_sweep.png` NOTES addendums (b) and (c).

E6 Mask dilation — Tested whether expanding the F200 mask further reduces σ. Negative finding: dilation inflates σ via noise — the dropped spaxels are S/N contributors, not bias contributors. Memory: `~/.claude/.../project_nb07d_overmasking_finding.md`. Notebook: `nb07d`.

E7 Arc-spectrum subtraction sibling — Build an arc-only spectrum from F200-masked spaxels, fit $\alpha\times$ arc + $\beta\times$ deflector model, subtract. Result matches §6cum within 0.1 km/s $\rightarrow$ *residual* arc dilution (the small fraction that leaks through the F200 mask) is sub-dominant at production statistics. The F200 mask already does the heavy lifting. Note: this test compares “masked + arc-subtract” against “masked alone” — it does NOT directly validate the no-mask vs masked Δ mechanism (that’s E8). Notebook: `nb07e`.

E8 Arc-as-sky mechanism test (rigorous test of the no-mask Δ) — Run §6cum at the *no-mask* R<R_e aperture with the bright outer-arc spectrum passed to ppxf as a `sky` template (free-amplitude additive component, NOT convolved with the deflector LOSVD — physically appropriate since the arc is at z=1.302, decoupled from the deflector kinematics). At N=500, FSPS+EMILES+XSL pooled:

Track	σ_e [km/s]	Δ from σ_A
A: masked headline	267.82	—
B: no-mask	252.82	-15.00
D: no-mask + arc-sky (E8)	274.64	+6.82

Recovery fraction $(\sigma\_D - \sigma\_B)/(\sigma\_A - \sigma\_B) = 145\%$ — i.e. arc-sky overshoots the masked headline by ~7 km/s. Interpretation:

- Continuum dilution explains the FULL no-mask Δ. There is no detectable kinematic-blend or template-mix mechanism contributing in the *opposite* direction.
- The 7 km/s overshoot is consistent with the known caveat that the outer-arc-mask spaxels (R > 3R_e/4) still receive seeing-blurred deflector light at the few-percent level, so subtracting $\alpha \times$ arc oversubtracts a small amount of *real* deflector flux too. This inflates σ slightly.
- The masked headline (Track A, paper number) remains the cleaner measurement because it doesn't have this oversubtraction artefact.
- α_arc $\approx$ 0.31 across all three SPS — internally consistent.

Code: `scripts/run_07f_arc_sky.py` (uses `setup_ppxf_inputs_from_spectrum` + ppxf `sky=` argument). Display: `notebooks/07f_arc_sky_subtract.ipynb`. Caches: `results/arc_sky_07f/{sps}_N500.npz` + `results/arc_sky_07f.npz`. Figures: `results/figures/nb07f_{arc_template,recovery,posteriors}.png`.

F. Centering

F1 5-center sweep — Tested HST_mean (paper choice), F140W only, F200LP only, IFU peak, arc-suppressed IFU peak. σ_e spread = 3.7 km/s. Carried as $\pm$4 km/s in the error budget. Notes: `NOTES_centering_investigation_2026-04-27.md`.

F2 F140W vs F200LP centroid — Δ = 0.36" — driven by the F200LP arc dragging that filter's centroid toward the arc. F140W is the cleaner bulge centroid; HST-mean is the robust compromise.

G. Bootstrap and polynomial degree

G1 Wild bootstrap — Hybrid Rademacher sign-flip $\times$ local-residual scaling (75-pix rolling window). For each iteration: residual = (galaxy − bestfit) $\times$ sign, scaled by local σ. Re-fit, record V/σ. Implemented in `scripts/bootstrap_ppxf.py:run_bootstrap_single_degree`.

G2 Parallel bootstrap — joblib + threadpoolctl pinning BLAS=1 per worker (otherwise BLAS thread oversubscription pathology — saw 4-hour runtimes). 8 workers gives ~6 min for 6 fits at N=500. Script: `scripts/bootstrap_ppxf_parallel.py:run_bootstrap_single_degree_parallel`.

G3 Polynomial degree sweep — Degrees 15-29 (15 values) per fit. Diagnostic: σ vs degree should be flat within bootstrap envelope; monotonic trend indicates the polynomial is absorbing LOSVD signal. Plotted at `nb09 §6.5` and saved to `results/figures/nb09_sigma_vs_degree.png`. Saturation confirmed at this degree range — picked from earlier `notebooks/03c` analysis.

G4 N=50 vs N=500 — N=50 smoke and N=500 production agree on σ_e to within 1 km/s. N=50 is ~6 min, N=500 is ~50 min. Production headline is N=500; the smoke is for quick rebuilds.

H. Cross-checks

H1 §7 discrete Gültekin (refreshed 2026-05-01) — `notebooks/07c §7`, recomputed with frame-aware ppxf via `scripts/refresh_07c_gultekin.py`. Annular sum $\sigma\_e^2(<R) = \Sigma\, F\_j\, (V\_j^2 + \sigma\_j^2)\, /\, \Sigma\, F\_j$ with arc-filter on outer annulus.

	Pre-frame-fix (Apr 24)	Post-frame-fix (May 1)	Δ
σ_e (arc-filtered)	254.99 −24.2/+28.4	256.17 −13.0/+12.7	+1.18

$\Delta < \pm 5$ km/s frame budget. Errors halved because the V_sys split collapsed from ~99 km/s (pre-fix) to ~8 km/s (post-fix per-SPS V_sys: fsps −11.2, emiles −7.4, xsl −3.1), shrinking the V^2 contribution to the Gültekin sum's MC scatter.

H2 §7b flat-σ extrapolation (refreshed 2026-05-01) — `notebooks/07c §7b`, same refresh. Push the §7 sum into the outer flagged bin assuming σ_outer = σ at the last clean annulus.

	Pre-frame-fix	Post-frame-fix	Δ
σ_e	271 −33/+35	274.37 −16.2/+17.4	+3.37

Both H1 and H2 remain consistent with the §6cum headline (267.82) at <1σ. The point of these cross-checks is preserved by the refresh, and the post-fix bands are tighter, strengthening rather than weakening the consistency.

Note: nb07e (E7, arc-spectrum-subtraction sibling) was *not* refreshed because its result is a relative comparison (matches §6cum within 0.1 km/s) — a uniform per-SPS σ shift moves both arms of the comparison together and the relative match is unchanged.

H3 F200 mask sensitivity (independent) — `results/annular_bootstrap_07c_nomask/`. Same §6cum pipeline with mask off. σ_e(<R_e) = 250.96 km/s. $\Delta = -16.4$ km/s. Independently confirms the nb09 mask-weight sweep at w=1.0.

I. Earlier sigma-discrepancy work (**`notebooks/05x`**)

The 21-test diagnostic notebook resolving the $\sigma \approx 301$ km/s (190-spaxel box, NB01) vs $\sigma \approx 210$ km/s (inner circular aperture, NB05) discrepancy. Resolution: σ depends on aperture; the 190-spaxel box is contaminated by the lensed arc. This work motivated the cumulative-I path (Tests 7-13 explored masking variants, Tests 14-16 went radial, Tests 18-19 tried binning approaches). PowerBin (Test 19) was rejected. The headline analysis moved to circular apertures + cumulative I-weighting (nb06 $\rightarrow$ nb07c $\rightarrow$ nb09).

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3. File index

Documents (top-level)

File	Purpose
`TESTS_AND_DIAGNOSTICS.md`	This file — master test index
`NOTES_nb09_frame_fix_and_final_sigma_e_2026-04-28.md`	Frame fix derivation + addenda (a)–(d)
`HANDOVER.md`	Apr 27 σ_e analysis snapshot (pre-frame-fix; superseded by nb09)
`HANDOFF_Ir_sweep_2026-04-28.md`	I(r) sweep design doc (consumed; sweep complete)
`NOTES_methodology_2026-04-27.md`	Cube provenance + earlier methodology
`NOTES_centering_investigation_2026-04-27.md`	5-center sweep result + $\pm$4 km/s budget
`CLAUDE.md`	Repo conventions, key results summary
`README.md`	High-level project description

Notebooks (paper-ready / current)

Notebook	Purpose
`notebooks/09_final_sigma_e_paper.ipynb`	Paper-ready σ_e (headline 268 $\pm$ 32 km/s)
`notebooks/04_redshift_verification.ipynb`	z = 0.67564 line-fit
`notebooks/02_streamlined_Bagpipes_SED.ipynb`	log M$\star$ = 11.33
`notebooks/08_sersic_total_photometry.ipynb`	Sersic-total photometry cross-check
`notebooks/07c_sigma_e_equalN.ipynb`	§6cum + §7 equal-N annular cross-check
`notebooks/07e_sigma_e_arc_subtract.ipynb`	Arc-subtraction sibling cross-check
`notebooks/05x_sigma_discrepancy_diagnostic.ipynb`	Tests 1-21 (foundational diagnostics)

Notebooks (history / reference, not paper-driving)

`01_*`, `02_*`, `03_bootstrap_ppxf_errors`, `03b_*`, `03c_*`, `05_*`, `06_*`, `07_*`, `07a_*`, `07b_*`, `07d_*` — kept for the history of the analysis.

Scripts (production)

Script	Purpose
`scripts/final_sigma_e.py`	nb09 production driver — runs all 3 mask tracks at N=500
`scripts/build_nb09.py`	Generates the paper notebook from the saved artifacts
`scripts/bootstrap_ppxf.py`	Frame-aware ppxf input prep + wild bootstrap
`scripts/bootstrap_ppxf_parallel.py`	joblib parallel bootstrap (BLAS=1 per worker)
`scripts/measure_Re.py`	R_e via masked CoG (multi-band, multi-strategy)
`scripts/redshift_verify.py`	Line-fitting redshift module
`scripts/ifu_spectral_resolution.py`	LSF audit

Scripts (sweeps and audits)

Script	Purpose
`scripts/audit_ppxf_methodology.py`	4 orthogonal correctness audits
`scripts/sigma_inst_sensitivity.py`	DISPSCAL × {0.5..2.0} sweep
`scripts/run_isource_shape_sweep.py`	10-shape I(r) sweep
`scripts/analyze_isource_shape_sweep.py`	I-shape sweep analysis + figure
`scripts/soft_mask_track.py`	Mask-weight runs (CLI `--weight`)
`scripts/analyze_mask_weight_sweep.py`	5-point sweep analysis + linearity fit
`scripts/regen_s6cum_nomask_diagnostic.py`	Centering investigation
`scripts/relock_nomask_Re_N500.py`	One-off N=500 relock for the no-mask track

Result files (key)

File	Contents
`results/final_sigma_e_paper.npz`	nb09 headline + per-SPS + 3-track summaries
`results/final_sigma_e_paper/`	Per-(aperture, SPS, mask_weight) caches
`results/sigma_e_ishape_sweep.npz`	I-shape sweep summary
`results/sigma_e_ishape_sweep_oldframe.npz`	Pre-frame-fix I-shape (audit)
`results/annular_bootstrap_07c_ishape/`	Per-(shape, SPS) I-shape caches (N=500)
`results/annular_bootstrap_07c_ishape_oldframe/`	Pre-frame-fix I-shape caches (N=250)
`results/annular_bootstrap_07c_nomask/`	§6cum nomask caches
`results/mask_weight_sweep.npz`	5-point mask-weight sweep summary
`results/ppxf_methodology_audit.npz`	4-audit results
`results/figures/nb09_*.png`	nb09 paper figures
`results/figures/nb07c_ishape_sweep.png`	I-shape sweep figure

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4. Reproduction recipes

```
conda activate ISMgas

# === Headline pipeline (~50 min at N=500) ===
python scripts/final_sigma_e.py --n_bootstrap 500
python scripts/build_nb09.py
jupyter nbconvert --to notebook --execute --inplace notebooks/09_final_sigma_e_paper.ipynb

# === Methodology audits (~3 min) ===
python scripts/audit_ppxf_methodology.py
python scripts/sigma_inst_sensitivity.py

# === I-shape sweep (~50 min at N=500) ===
python scripts/run_isource_shape_sweep.py --n-bootstrap 500 --n-jobs 8
python scripts/analyze_isource_shape_sweep.py

# === Mask-weight sweep (~25 min total) ===
# (w=0 and w=1 are produced by final_sigma_e.py; only need 0.25/0.5/0.75)
python scripts/soft_mask_track.py --weight 0.25
python scripts/soft_mask_track.py --weight 0.50
python scripts/soft_mask_track.py --weight 0.75
python scripts/analyze_mask_weight_sweep.py

# === Smoke run (~6 min, useful for rebuilds) ===
python scripts/final_sigma_e.py --n_bootstrap 50
```

Caches are keyed by N_bootstrap and mask_weight in the filename — re-runs at the same N skip cached fits unless `--force` is passed.

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5. What's intentionally NOT done

- Re-run nb09 at higher N (e.g. N=2000): the pooled posterior is already smooth; SPS systematics dominate the residual statistical variance.
- Re-fit Sersic2D F200LP: the n=0.30 fit is non-physical due to UV-arc domination; the F140W Sersic fit is the trustworthy one. F200LP_Sersic2D is the I-shape outlier *by design* and is excluded from the budget.
- Voronoi binning — Test 18 / nb05x — abandoned; not pursued at N=500.
- PowerBin spatial binning — Test 19 / nb05x — rejected (outer-bin σ > 800 km/s); not pursued.
- Per-spaxel rotation map at production statistics — no rotation detected at the seeing limit; not paper-driving.
- Full nb07a Sersic-I revisit at the post-frame-fix headline — superseded by the I-shape sweep (D3/D4) which covers the Sersic2D shape.

See `~/.claude/projects/.../memory/MEMORY.md` for compressed cross-session references, including: - `reference_cumulative_vs_annular_sigma_e.md` — §6cum vs §7 design choice - `reference_sps_systematic.md` — per-SPS V_sys + frame-fix details - `reference_ppxf_vacair_handling.md` — vac/air conversion methodology - `reference_kcwi_data_properties.md` — KCWI cube + LSF + paper boilerplate - `project_nb09_final_sigma_e.md` — current headline status